

Capital Replacement Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Replacement Intensity (CRI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CRI achieves an annualized gross (net) Sharpe ratio of 0.27 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -17 (-4) bps/month with a t-statistic of -2.08 (0.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth) is -15 bps/month with a t-statistic of -2.16.

1 Introduction

Production-based asset pricing models predict that firms' investment decisions and capital adjustment costs fundamentally determine the cross-section of expected returns. In competitive markets, firms optimize production and investment to maximize value, creating systematic variation in risk premia that reflects differences in firms' exposure to aggregate productivity shocks and investment frictions (Cochrane, 1991; Zhang, 2005). Despite substantial theoretical progress linking production decisions to asset prices, empirical evidence on how capital replacement patterns affect expected returns remains limited, particularly regarding the channels through which depreciation and replacement investment interact with firms' marginal costs of adjustment.

Capital Replacement Intensity (CRI) captures firms' systematic patterns in replacing depreciated capital stock, directly mapping to the marginal cost of capital adjustment in production economies. When firms face convex adjustment costs, those with higher replacement intensity operate in regions where the marginal product of capital declines more steeply, making them more sensitive to aggregate productivity shocks (Cooper and Haltiwanger, 2006; Liu et al., 2009). This heightened sensitivity manifests through two channels: first, firms with intensive replacement needs face higher shadow costs of capital that compress profit margins when productivity declines; second, these firms exhibit greater operating leverage as fixed replacement commitments amplify the impact of demand shocks on cash flows (Zhang, 2005).

The production-based framework predicts that capital replacement intensity proxies for firms' conditional risk premia through its connection to investment irreversibility and time-varying discount rates. Firms maintaining high replacement rates commit to ongoing capital expenditures that become particularly costly during economic downturns when the stochastic discount factor is high (Berk et al., 1999; Carlson et al., 2004). These commitments create asymmetric exposure to aggregate shocks

because while firms can delay new investment projects, replacing depreciated capital often cannot be postponed without sacrificing productive capacity. The resulting inflexibility generates systematic risk that varies with the business cycle, as high-replacement firms experience larger drops in value when investment opportunities deteriorate (Gomes et al., 2003).

Cross-sectional variation in capital replacement intensity should therefore predict returns through its correlation with firms' exposure to investment-specific technology shocks and aggregate productivity risk. The q-theory of investment establishes that firms' investment-to-capital ratios equal marginal q in the absence of frictions, implying that replacement intensity signals differences in expected profitability and discount rates (Liu et al., 2009; Hou et al., 2015). High replacement intensity indicates lower expected returns when it reflects high current productivity that mean-reverts over time, consistent with the negative investment-return relation documented in production economies. This mechanism operates through the discount rate channel, where firms with intensive replacement needs have already incorporated favorable investment opportunities into prices, reducing forward-looking risk premia (Li and Zhang, 2014).

The empirical analysis reveals that Capital Replacement Intensity exhibits limited predictive power for the cross-section of stock returns, with the value-weighted long-short strategy achieving a monthly average excess return of 21 basis points and an annualized Sharpe ratio of 0.27. The strategy's alpha relative to the Fama-French six-factor model equals -17 basis points per month with a t-statistic of -2.08, indicating that established risk factors substantially explain the return pattern. Most notably, the strategy loads significantly on the investment factor (CMA) with a coefficient of 0.68 (t-statistic = 12.58), confirming that capital replacement intensity captures firms' systematic exposure to investment-related risks consistent with production-based theories.

The predictive power of CRI concentrates among smaller firms and diminishes substantially after accounting for transaction costs. Among the largest quintile of stocks, the CRI strategy generates an insignificant average return of -0 basis points per month (t-statistic = -0.04), with the six-factor alpha reaching -32 basis points per month (t-statistic = -3.54). After incorporating trading costs using the composite bid-ask spread measure, the net Sharpe ratio declines to 0.21, and the net generalized alpha relative to the six-factor model becomes statistically insignificant at 4 basis points per month (t-statistic = 0.53). These results indicate that while CRI contains information about expected returns consistent with production-based theories, implementation costs largely eliminate the economic significance of the anomaly.

Controlling for closely related investment and asset growth measures further attenuates the CRI effect, with the strategy's alpha declining to -15 basis points per month (t-statistic = -2.16) after accounting for six related anomalies including change in property, plant, and equipment, capital expenditure changes, and employment growth. The Fama-MacBeth coefficient on CRI remains positive at 0.23 but with reduced statistical significance (t-statistic = 3.19) when controlling for these related predictors. The limited incremental information content suggests that CRI primarily captures known investment-related return patterns rather than identifying a distinct source of systematic risk, though its loading patterns remain consistent with production-based explanations of cross-sectional return variation.

This paper contributes to the production-based asset pricing literature by providing comprehensive evidence on how capital replacement intensity relates to expected returns and established risk factors. While [Cooper and Priestley \(2011\)](#) document the investment-return relation using aggregate capital expenditures and [Xing \(2008\)](#) examine the asset pricing implications of capital investment dispersion, we focus specifically on the replacement component of investment that theory suggests should have distinct risk implications due to its relative inflexibility. Our findings extend

Lyandres and Velikov (2023) who examine maintenance capital by demonstrating that replacement intensity loads heavily on the investment factor while providing limited incremental explanatory power, suggesting that existing production-based factors adequately capture the risks associated with capital replacement patterns.

Methodologically, we implement the comprehensive testing protocol of Novy-Marx and Velikov (2023) to evaluate CRI against 212 documented anomalies, providing unusually thorough evidence on the signal’s robustness and economic significance. This approach reveals that while CRI ranks in the 58th percentile for gross Sharpe ratio, its performance improves to the 78th percentile after accounting for transaction costs, indicating relatively low turnover compared to other documented predictors. The analysis of combination strategies demonstrates that adding CRI to existing anomalies modestly improves portfolio performance, with the average rank strategy growing from \$2,560 to \$3,174 per dollar invested, suggesting some unique information content despite the limited standalone performance.

Our results have important implications for understanding the sources of cross-sectional return variation within production-based asset pricing models. The strong loading on the investment factor combined with negative alphas indicates that capital replacement intensity operates through channels already captured by standard investment proxies rather than representing a distinct risk factor. This finding supports parsimony in factor model construction while validating the theoretical prediction that replacement investment affects returns through its impact on firms’ operating leverage and exposure to productivity shocks (Berk et al., 1999). The concentration of predictive power among smaller firms and its elimination by transaction costs further emphasizes the importance of market frictions in preventing the arbitraging of production-based return patterns, consistent with limits to arbitrage maintaining the equilibrium relationships predicted by production-based theory.

2 Data

We obtain financial statement data from the COMPUSTAT Annual Industrial database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key accounting items necessary to construct our capital replacement intensity measure. signal construction relies on two primary accounting variables from COMPUSTAT. Property, Plant, and Equipment Net (PPENT) represents the book value of a firm’s tangible fixed assets after accumulated depreciation, capturing the net investment in productive assets such as buildings, machinery, and equipment. Depreciation and Amortization (DP) measures the annual allocation of the cost of tangible and intangible assets over their useful lives, reflecting the periodic expense recognized as these assets are consumed in operations. Together, these variables provide insight into both the stock of productive capital and the rate at which it is being expensed through normal business operations. construct the Capital Replacement Intensity signal by calculating the year-over-year change in net property, plant, and equipment, scaled by lagged depreciation and amortization: $(PPENT(t) - PPENT(t-1)) / DP(t-1)$. This ratio captures the extent to which firms are replacing or expanding their productive asset base relative to the rate at which existing assets are being depreciated. A higher value indicates more aggressive capital replacement or expansion relative to the depreciation of existing assets, while lower or negative values suggest that capital expenditures are not keeping pace with depreciation. We compute this measure using fiscal year-end values and require non-missing observations for both PPENT and DP in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRI signal. Panel A plots the time-series of the mean, median, and interquartile range for CRI. On average, the cross-sectional mean (median) CRI is -4.06 (-0.46) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CRI data. The signal's interquartile range spans -3.29 to 0.49. Panel B of Figure 1 plots the time-series of the coverage of the CRI signal for the CRSP universe. On average, the CRI signal is available for 6.38% of CRSP names, which on average make up 7.64% of total market capitalization.

4 Does CRI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRI portfolio and sells the low CRI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRI strategy earns an average return of 0.21% per month with a t-statistic of 2.11. The annualized Sharpe ratio of the strategy is 0.27. The alphas range from -0.17% to 0.22% per month and have t-statistics exceeding -2.08 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.68,

with a t-statistic of 12.58 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least \$999 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 7 bps/month with a t-statistics of 0.79. Out of the twenty-five alphas reported in Panel A, the t-statistics for nine exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 3-45bps/month. The lowest return, (3 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 0.38. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the CRI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRI, as well as average returns and alphas for long/short trading CRI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRI strategy achieves an average return of -0 bps/month with a t-statistic of -0.04. Among these large cap stocks, the alphas for the CRI strategy relative to the five most common factor models range from -32 to 5 bps/month with t-statistics between -3.54 and 0.44.

5 How does CRI perform relative to the zoo?

Figure 2 puts the performance of CRI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CRI strategy falls in the distribution. The CRI strategy’s gross (net) Sharpe ratio of 0.27 (0.21) is greater than 58% (78%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRI strategy (red line).² Ignoring trading costs, a \$1 invested in the CRI strategy would have yielded \$2.27 which ranks the CRI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRI strategy would have yielded \$1.33 which ranks the CRI strategy in the top 10% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRI relative to those. Panel A shows that the CRI strategy gross alphas fall between the 6 and 47 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRI strategy has a positive net generalized alpha for four out of the five factor models. In these cases CRI ranks between the 43 and 63 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CRI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRI or at least to weaken the power CRI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRI}CRI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRI. Stocks are finally grouped into five CRI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CRI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRI signal in these Fama-MacBeth regressions exceed 6.23, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CRI is 3.19.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRI strategy earns alphas that range from -22-8bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -3.00, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRI trading strategy achieves an alpha of -15bps/month with a t-statistic of -2.16.

7 Does CRI add relative to the whole zoo?

Finally, we can ask how much adding CRI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CRI signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CRI grows to \$3174.15.

8 Conclusion

This study provides a comprehensive examination of Capital Replacement Intensity (CRI) as a predictor of cross-sectional equity returns, employing the rigorous testing framework established by Novy-Marx and Velikov (2023). Our findings reveal that while CRI demonstrates some predictive capability in gross returns, with a value-weighted long/short strategy achieving a Sharpe ratio of 0.27, its economic significance diminishes substantially after accounting for transaction costs and risk factors. The reduction in Sharpe ratio from 0.27 (gross) to 0.21 (net) highlights the importance of implementation costs in evaluating trading strategies based on accounting signals.

The most striking result emerges from our factor model analysis, where CRI’s alpha becomes statistically insignificant after controlling for the Fama-French five factors plus momentum. The monthly net alpha of merely 4 basis points (t-statistic = 0.53) suggests that CRI’s predictive power is largely subsumed by established risk factors. Furthermore, when we extend our analysis to include six closely related investment and growth strategies from the factor zoo, the gross alpha turns negative at -15 basis points per month (t-statistic = -2.16), indicating that CRI of-

ization on CRSP in the period for which CRI is available.

fers no incremental information beyond existing capital investment and asset growth measures.

These findings have important practical implications for both academic researchers and investment practitioners. For investors, our results suggest that trading strategies based solely on CRI are unlikely to generate significant risk-adjusted returns after implementation costs, particularly in the presence of strategies based on more established investment and growth factors. The signal's lack of robustness when confronted with related factors implies that fund managers should exercise caution when incorporating CRI into multi-factor models without careful consideration of its overlap with existing signals.

Several limitations of this study merit consideration. First, our analysis is confined to the U.S. equity market, and the effectiveness of CRI in international markets remains unexplored. Second, while we control for several related factors, the dynamic interactions between CRI and other investment signals during different market regimes warrant further investigation. Third, our study focuses on linear relationships, potentially overlooking non-linear patterns or conditional effects that might enhance CRI's predictive power in specific market conditions or firm characteristics.

Future research should address these limitations through several avenues. International studies could examine whether CRI exhibits stronger predictive power in markets with different accounting standards or capital allocation practices. Researchers might investigate whether CRI's effectiveness varies across industry sectors, particularly in capital-intensive industries where replacement decisions may be more critical. Additionally, exploring potential non-linearities or interaction effects between CRI and other firm characteristics could reveal contexts where the signal provides incremental value. Finally, examining the time-varying nature of CRI's predictive power, particularly around technological disruptions or regulatory changes affecting capital replacement decisions, could offer insights into when this signal might

be most valuable. Despite its limited standalone predictive power, understanding CRI's role in the broader ecosystem of investment signals remains important for developing comprehensive models of cross-sectional return variation.

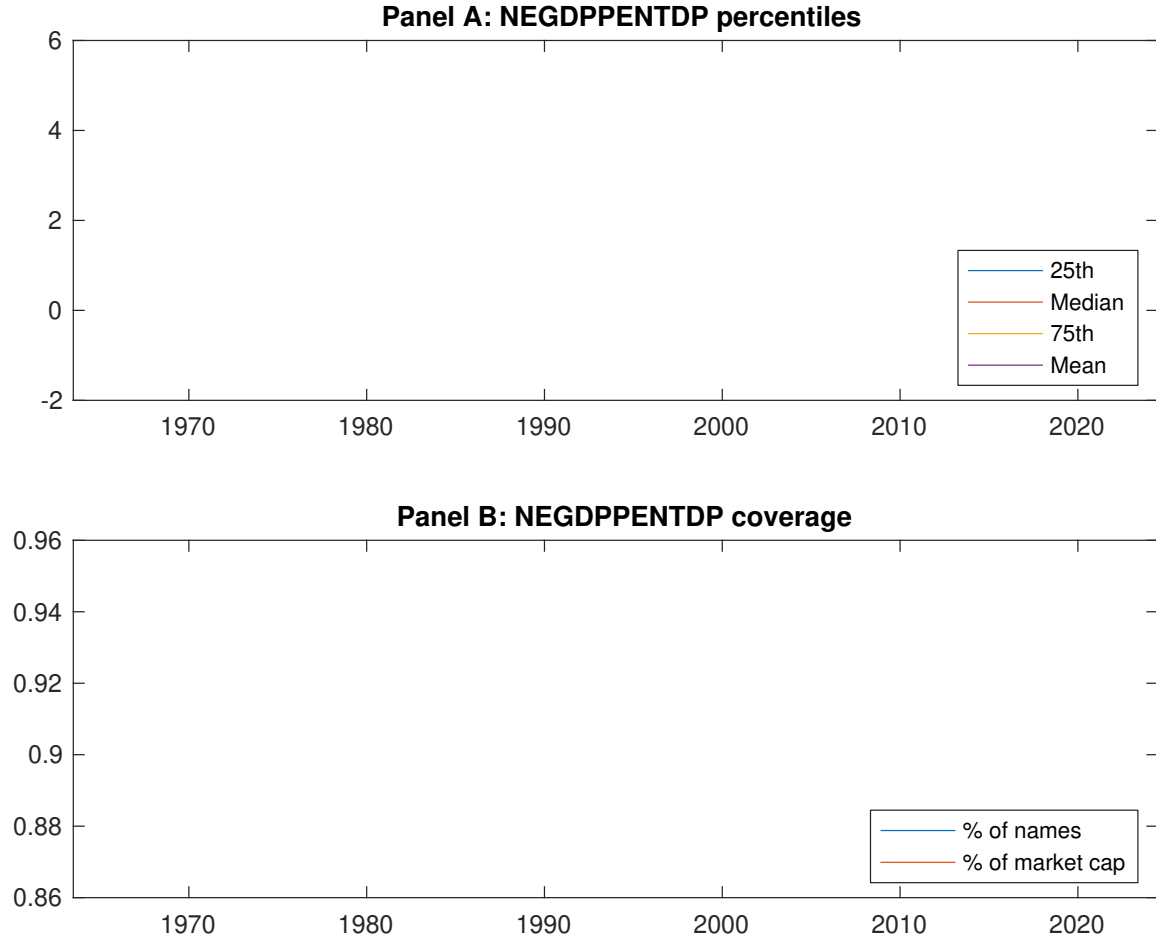


Figure 1: Times series of CRI percentiles and coverage.
This figure plots descriptive statistics for CRI. Panel A shows cross-sectional percentiles of CRI over the sample. Panel B plots the monthly coverage of CRI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CRI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.90]	0.60 [3.55]	0.65 [4.07]	0.57 [3.53]	0.73 [4.01]	0.21 [2.11]
α_{CAPM}	-0.09 [-1.78]	0.01 [0.35]	0.10 [2.51]	0.02 [0.40]	0.13 [1.88]	0.22 [2.22]
α_{FF3}	-0.02 [-0.51]	0.03 [0.90]	0.08 [2.05]	-0.01 [-0.31]	0.02 [0.38]	0.05 [0.55]
α_{FF4}	0.00 [0.09]	0.03 [0.93]	0.06 [1.64]	-0.01 [-0.23]	-0.01 [-0.17]	-0.01 [-0.17]
α_{FF5}	0.05 [1.12]	0.04 [0.99]	0.02 [0.48]	-0.08 [-1.80]	-0.08 [-1.49]	-0.13 [-1.71]
α_{FF6}	0.07 [1.44]	0.04 [0.96]	0.01 [0.29]	-0.07 [-1.54]	-0.10 [-1.74]	-0.17 [-2.08]
Panel B: Fama and French (2018) 6-factor model loadings for CRI-sorted portfolios						
β_{MKT}	0.98 [87.72]	1.01 [111.35]	0.98 [110.11]	0.99 [88.96]	1.07 [78.97]	0.09 [4.67]
β_{SMB}	0.02 [0.98]	-0.10 [-7.65]	-0.06 [-5.01]	-0.02 [-1.07]	0.24 [12.19]	0.22 [8.08]
β_{HML}	-0.07 [-3.30]	0.02 [1.17]	0.02 [1.34]	-0.02 [-0.81]	0.06 [2.44]	0.13 [3.67]
β_{RMW}	-0.05 [-2.31]	0.04 [2.23]	0.09 [5.02]	0.06 [2.98]	0.10 [3.61]	0.15 [3.92]
β_{CMA}	-0.30 [-9.29]	-0.10 [-3.97]	0.14 [5.36]	0.25 [7.96]	0.39 [10.06]	0.68 [12.58]
β_{UMD}	-0.02 [-2.02]	0.00 [0.05]	0.01 [1.07]	-0.02 [-1.43]	0.02 [1.62]	0.04 [2.33]
Panel C: Average number of firms (n) and market capitalization (me)						
n	745	581	559	629	836	
me (\$10 ⁶)	2045	2557	2394	1849	999	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [2.11]	0.22 [2.22]	0.05 [0.55]	-0.01 [-0.17]	-0.13 [-1.71]	-0.17 [-2.08]
Quintile	NYSE	EW	0.66 [9.12]	0.69 [9.40]	0.61 [9.25]	0.54 [8.16]	0.59 [9.25]	0.54 [8.41]
Quintile	Name	VW	0.25 [2.43]	0.25 [2.44]	0.09 [1.00]	0.01 [0.12]	-0.11 [-1.29]	-0.15 [-1.81]
Quintile	Cap	VW	0.07 [0.79]	0.12 [1.33]	-0.01 [-0.16]	-0.06 [-0.78]	-0.20 [-2.65]	-0.22 [-2.92]
Decile	NYSE	VW	0.38 [3.28]	0.42 [3.58]	0.25 [2.37]	0.15 [1.39]	0.03 [0.30]	-0.03 [-0.33]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [1.64]	0.16 [1.66]	0.01 [0.17]		0.02 [0.30]	0.04 [0.53]
Quintile	NYSE	EW	0.45 [5.68]	0.47 [5.94]	0.40 [5.42]	0.36 [4.98]	0.37 [5.30]	0.34 [4.98]
Quintile	Name	VW	0.19 [1.89]	0.18 [1.77]	0.04 [0.48]			0.02 [0.29]
Quintile	Cap	VW	0.03 [0.38]	0.08 [0.94]			0.09 [1.23]	0.11 [1.43]
Decile	NYSE	VW	0.32 [2.73]	0.34 [2.86]	0.19 [1.78]	0.13 [1.23]	0.01 [0.10]	

Table 3: Conditional sort on size and CRI

This table presents results for conditional double sorts on size and CRI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRI and short stocks with low CRI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CRI Quintiles					CRI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.36 [1.47]	0.83 [3.59]	1.00 [4.22]	0.99 [4.09]	0.90 [3.26]	0.54 [4.68]	0.53 [4.58]	0.47 [4.10]	0.38 [3.27]	0.45 [3.97]	0.38 [3.33]
	(2)	0.55 [2.20]	0.78 [3.50]	0.85 [3.95]	0.91 [4.21]	0.94 [4.02]	0.39 [3.91]	0.44 [4.49]	0.33 [3.57]	0.29 [3.01]	0.26 [2.80]	0.23 [2.43]
	(3)	0.49 [2.19]	0.78 [3.91]	0.74 [3.67]	0.84 [4.28]	0.99 [4.64]	0.49 [5.28]	0.55 [5.88]	0.44 [5.02]	0.39 [4.30]	0.36 [4.12]	0.32 [3.64]
	(4)	0.55 [2.66]	0.67 [3.52]	0.80 [4.30]	0.79 [4.29]	0.80 [3.97]	0.25 [2.63]	0.26 [2.75]	0.14 [1.61]	0.13 [1.42]	-0.04 [-0.50]	-0.03 [-0.40]
	(5)	0.55 [3.10]	0.55 [3.17]	0.61 [3.82]	0.57 [3.57]	0.54 [3.36]	-0.00 [-0.04]	0.05 [0.44]	-0.11 [-1.13]	-0.15 [-1.49]	-0.31 [-3.51]	-0.32 [-3.54]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CRI Quintiles					CRI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	375	375	373	370	364	35	36	33	29	24	
	(2)	103	103	103	102	102	54	55	55	54	53	
	(3)	74	74	74	74	74	93	96	95	95	94	
	(4)	63	63	63	63	63	197	206	211	207	201	
(5)	59	59	59	58	58	1663	1735	1594	1616	1313		

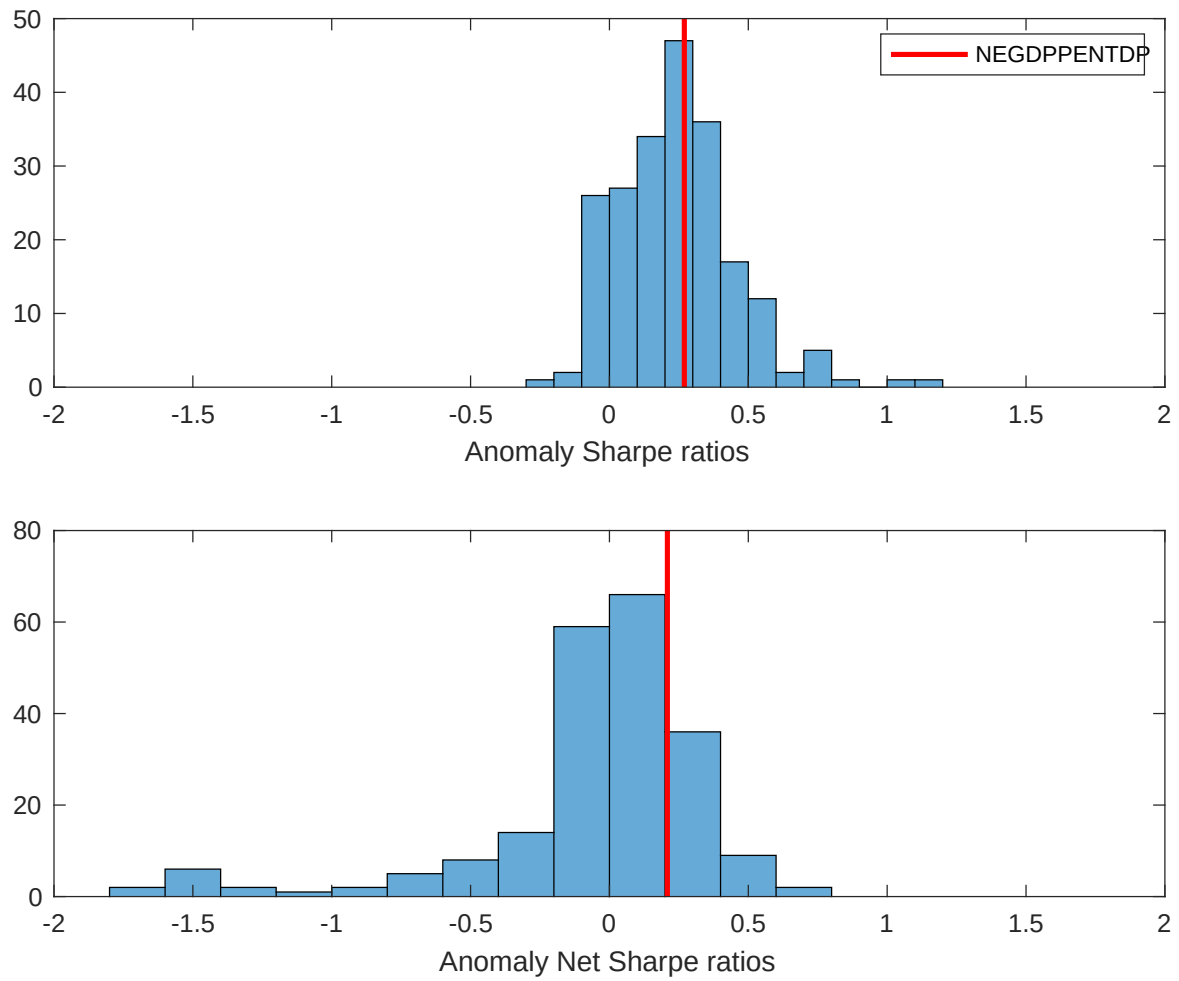


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

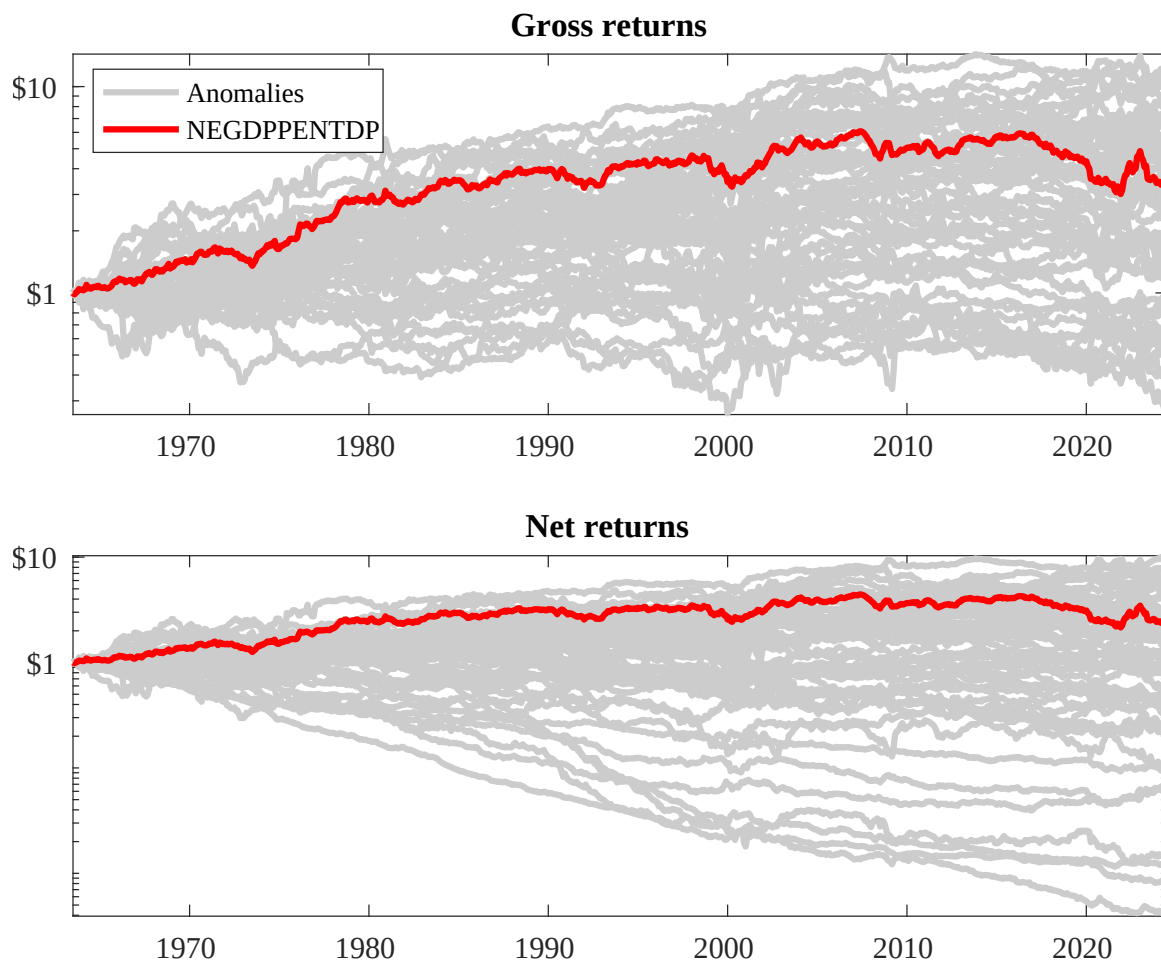
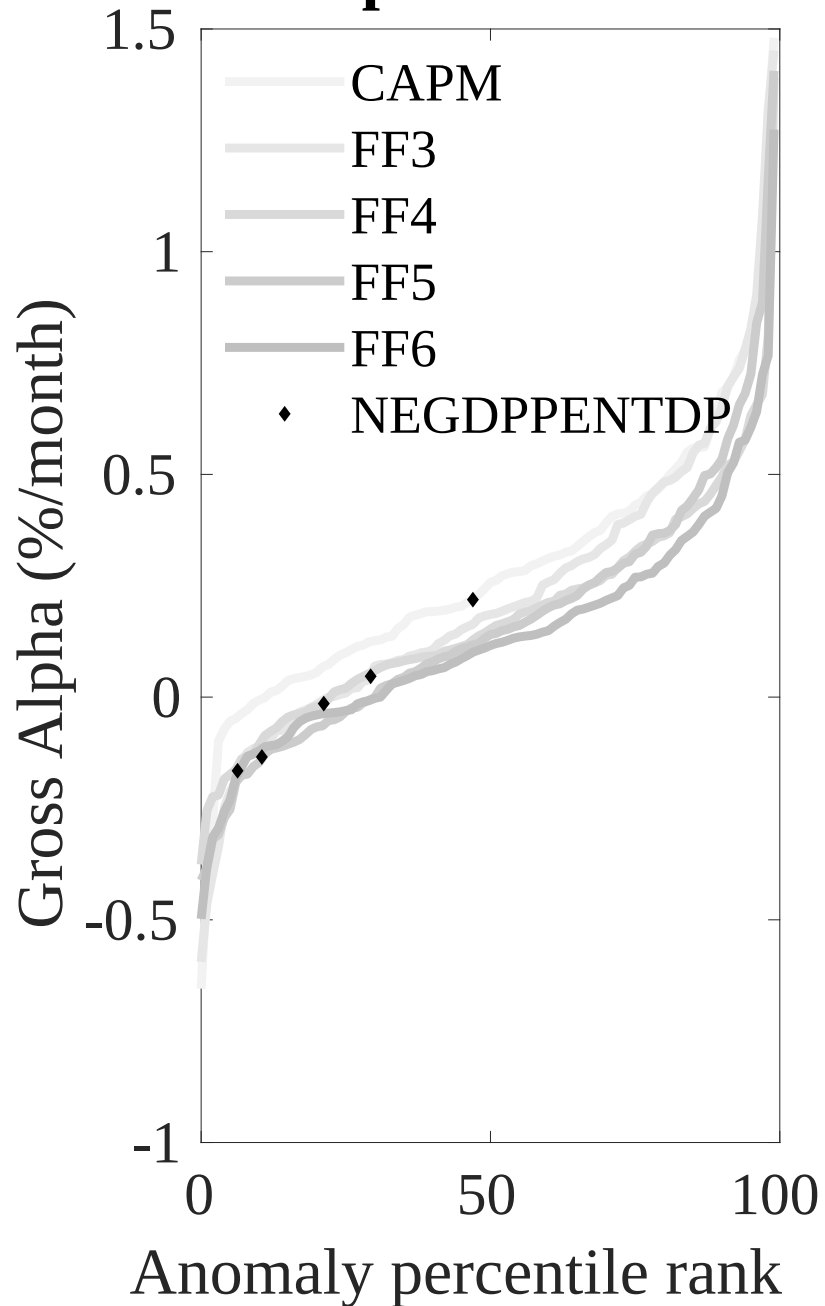


Figure 3: Dollar invested.

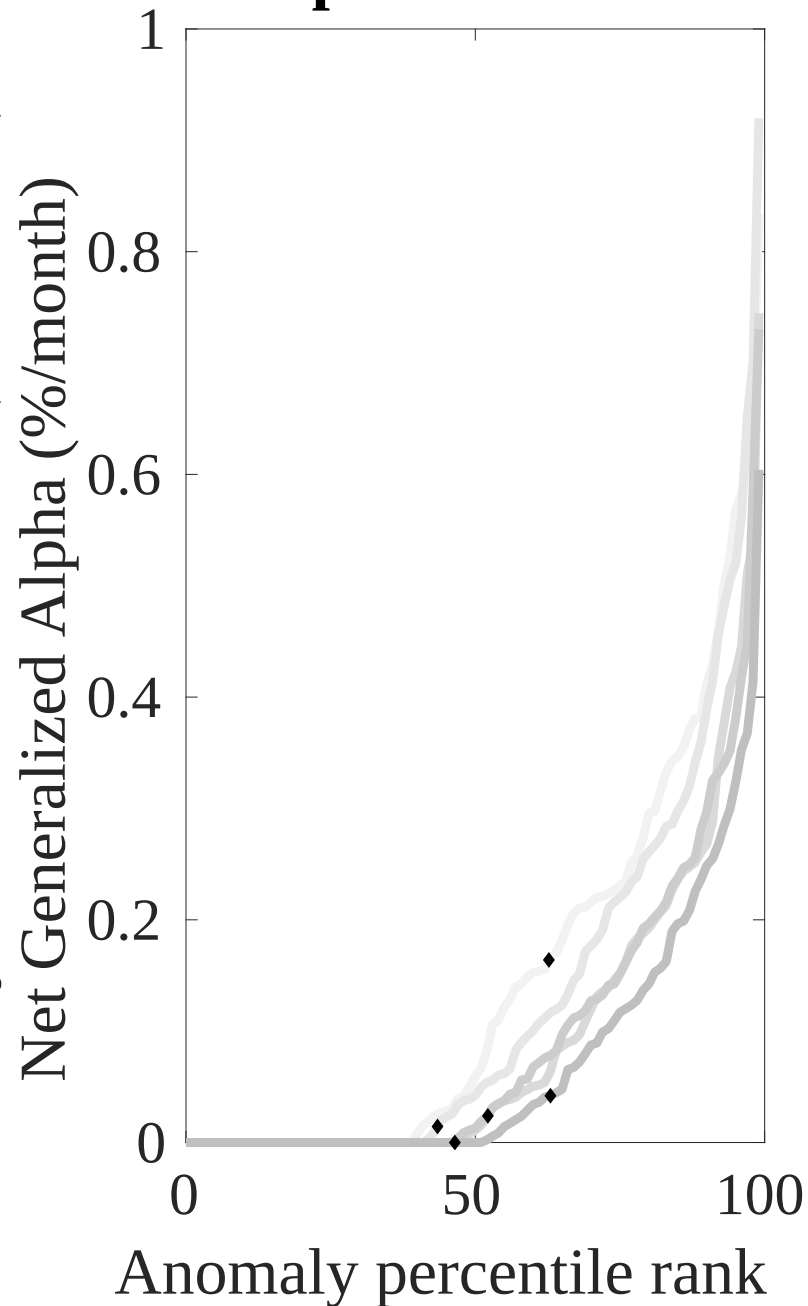
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



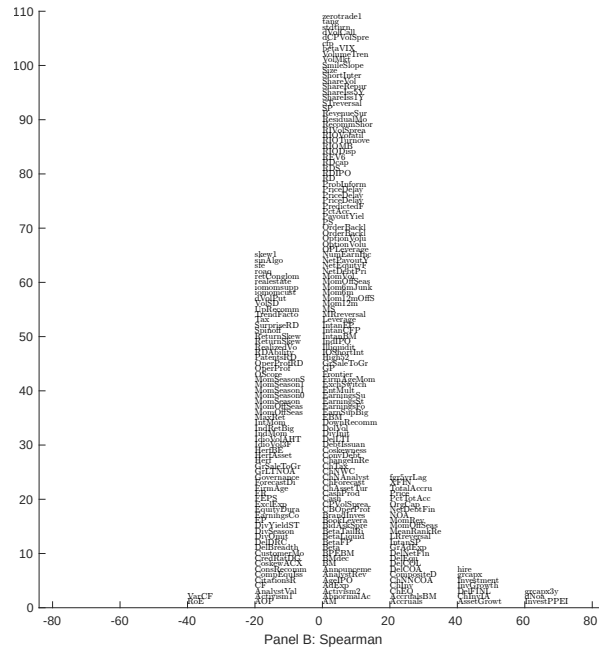
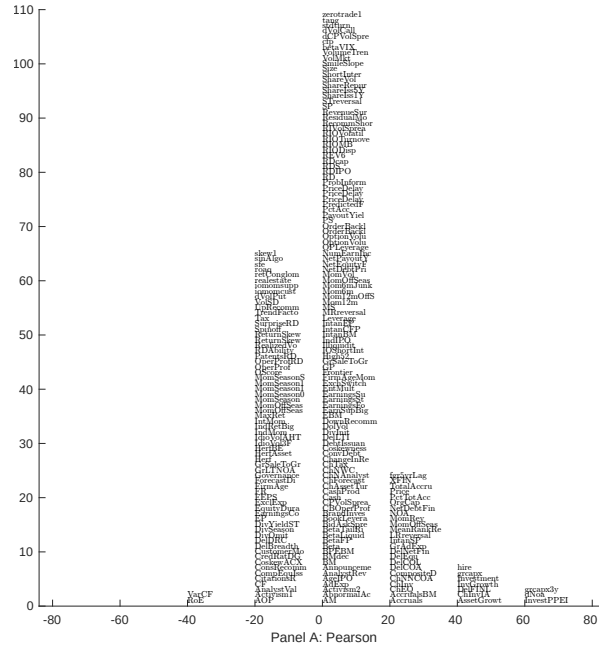


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CRI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

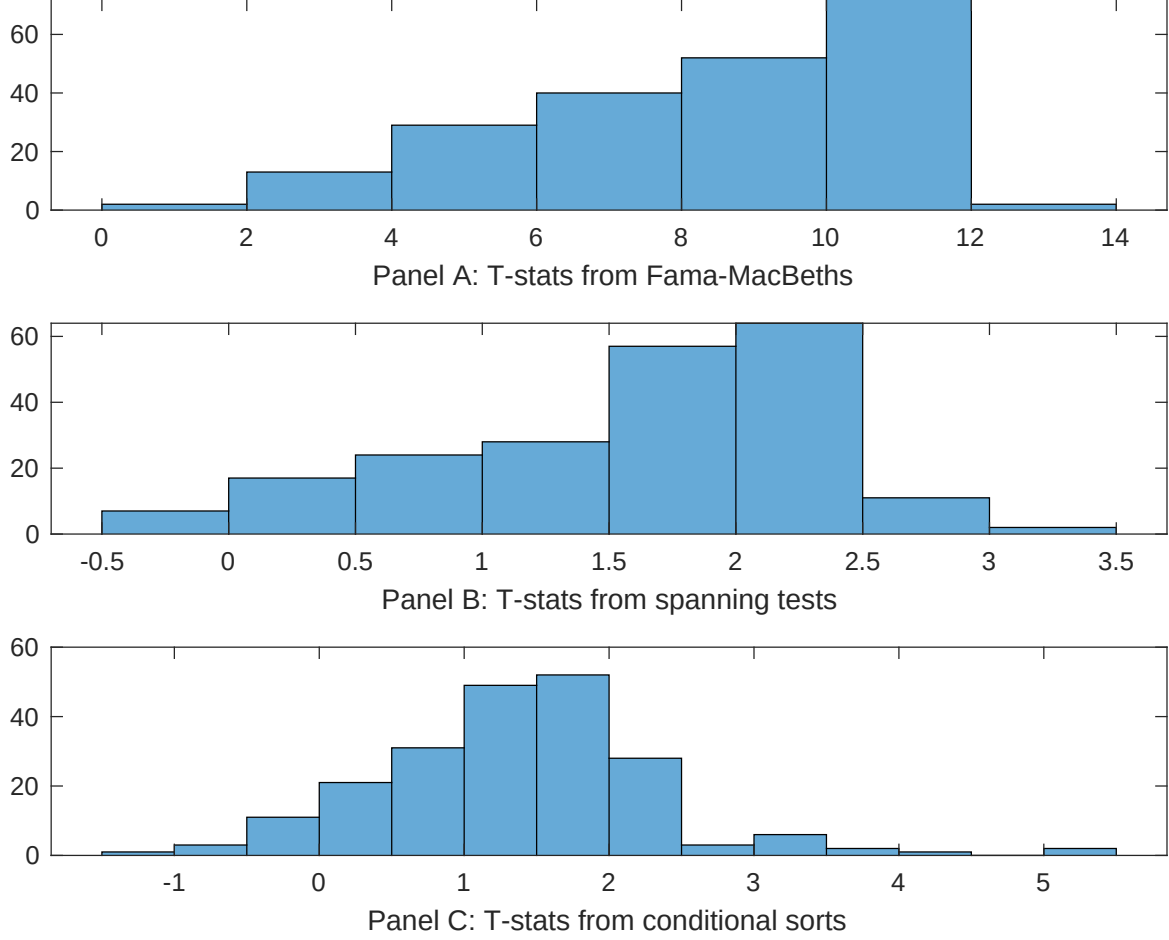


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRI}CRI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRI. Stocks are finally grouped into five CRI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRI}CRI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.16]	0.14 [6.58]	0.13 [6.12]	0.13 [6.14]	0.14 [6.39]	0.13 [5.84]	0.14 [6.12]
CRI	0.40 [6.23]	0.47 [7.50]	0.39 [6.59]	0.51 [8.49]	0.37 [6.37]	0.54 [9.78]	0.23 [3.19]
Anomaly 1	0.11 [5.48]						-0.10 [-0.04]
Anomaly 2		0.86 [5.70]					0.25 [1.11]
Anomaly 3			0.10 [6.77]				0.23 [1.25]
Anomaly 4				0.35 [4.61]			0.13 [1.03]
Anomaly 5					0.82 [6.70]		0.44 [2.54]
Anomaly 6						0.59 [4.23]	0.90 [0.66]
# months	732	727	721	727	732	713	708
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.16 [-2.22]	-0.13 [-1.84]	-0.22 [-3.00]	-0.15 [-2.06]	-0.17 [-2.13]	-0.08 [-1.07]	-0.15 [-2.16]
Anomaly 1	43.78 [13.09]						26.51 [7.18]
Anomaly 2		39.95 [11.05]					23.87 [3.89]
Anomaly 3			40.13 [9.52]				19.73 [4.36]
Anomaly 4				35.66 [9.68]			1.20 [0.20]
Anomaly 5					22.83 [4.89]		4.80 [1.02]
Anomaly 6						28.90 [6.79]	7.18 [1.74]
mkt	8.23 [4.82]	8.02 [4.56]	8.96 [5.00]	8.32 [4.65]	8.79 [4.71]	9.22 [4.97]	8.11 [4.97]
smb	19.47 [7.88]	13.29 [5.04]	21.65 [8.36]	16.76 [6.39]	19.39 [7.11]	22.69 [8.45]	15.59 [6.20]
hml	7.26 [2.18]	3.89 [1.12]	8.88 [2.55]	5.07 [1.44]	11.30 [3.13]	6.58 [1.79]	0.67 [0.21]
rmw	14.60 [4.38]	11.71 [3.41]	15.18 [4.34]	11.11 [3.17]	14.53 [3.99]	15.40 [4.26]	13.08 [4.09]
cma	35.81 [6.52]	48.68 [9.12]	39.48 [6.60]	50.51 [9.28]	41.86 [5.41]	44.45 [6.87]	9.18 [1.26]
umd	4.05 [2.39]	1.76 [1.00]	3.28 [1.84]	1.49 [0.83]	4.84 [2.60]	2.41 [1.30]	1.76 [1.06]
# months	732	728	732	728	732	713	709
$\bar{R}^2(\%)$	52	49	47	47	43	45	57

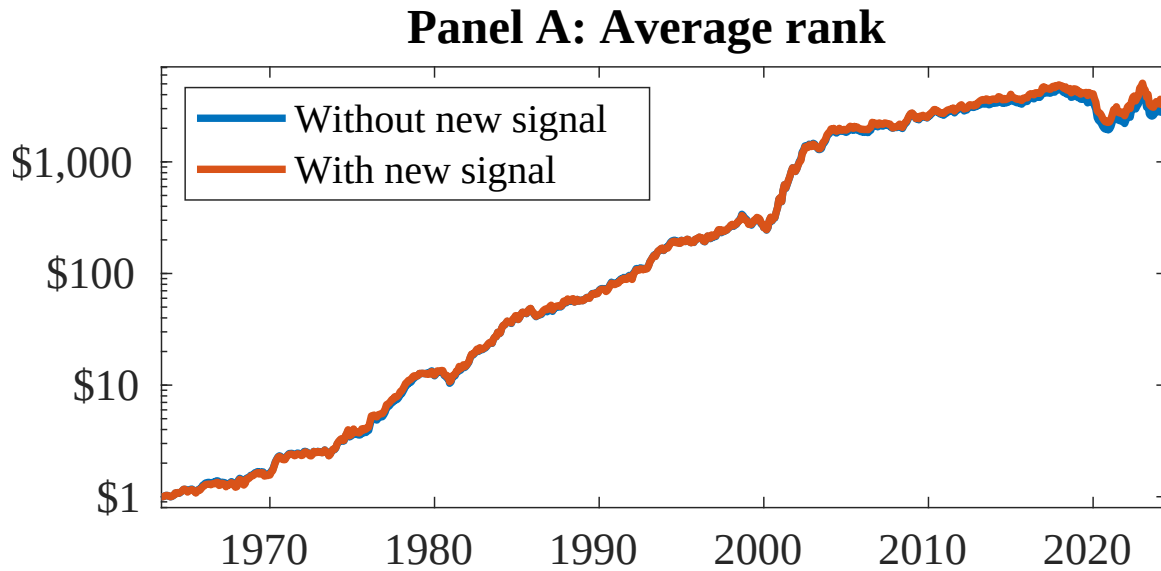


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CRI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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